Missing Data in Longitudinal Blood Pressure: GEE vs. Mixed Models

Missing Data Associations

Association between missing data and baseline predictors

Question: Using the wide format data, I examine the association between missing data at year1 (missing=1) with the predictors collected at the baseline visit. Are baseline factors associated with a missing blood pressure at year 1?

Analysis: - Missing data patterns: 1384 participants (53.1%) have missing=0, 1223 participants (46.9%) have missing=1

- **Baseline SBP (sbp0) association**:
- Missing=0: Mean = 122.2 mmHg (SD = 11.8), Median = 123 mmHg
- Missing=1: Mean = 149.2 mmHg (SD = 15.0), Median = 147 mmHg
- Mean difference = 26.9 mmHg, t-test $p < 0.001$
- Cohen's d = 2.01 (large effect size)
- Assumption violations: Non-normal distributions (Shapiro-Wilk $p < 0.001$), unequal variances (F-test $p < 0.001$)
- Outlier assessment: Minimal outliers detected using IQR method
- Sample size: n=1,384 (missing=0), n=1,223 (missing=1) provides excellent power (>99%) for large effect
- **Baseline BMI association**:
- Missing=0: Mean = 28.3, Missing=1: Mean = 28.9
- Mean difference = 0.53, t-test $p = 0.014$
- **Baseline diabetes association**:
- Missing=0: 22.0% diabetic, Missing=1: 30.5% diabetic
- Chi-square test: χ^2 $p < 0.001$, df = 1
- Odds Ratio = 1.55, 95% CI: (1.30, 1.85)

Summary: Baseline factors are associated with missing blood pressure at year 1. Participants with missing data have higher baseline SBP, slightly higher BMI, and higher rates of diabetes.

Year 1 SBP and missingness association

Question: Does year 1 SBP (sbp1) appear to be associated with a missing value of blood pressure at year 1 (missing)?

Analysis: Mean year 1 SBP is 127.5 mmHg for missing=0 and 143.9 mmHg for missing=1. Mean difference is 16.3 mmHg. T-test p-value < 0.001.

Answer: Yes, year 1 SBP is associated with missing status. Participants flagged as missing have higher year 1 SBP values.

Classify Missingness Scenarios (MCAR/MAR/NMAR)

Scenario (a): Missingness only depends on baseline SBP (sbp0)

Classification: MAR (Missing at Random) **Explanation**: To classify this scenario, we must determine whether baseline SBP represents observed or unobserved data. Since baseline SBP is measured and recorded for all participants, it constitutes observed data. The VGSM establishes that "data are said to be missing completely at random (MCAR) if P(Ri=1) does not depend on any of the variables" - but this scenario violates MCAR because missingness explicitly depends on baseline SBP values. However, van Buuren (*Flexible Imputation of Missing Data*) provides the key criterion: "If the probability of being missing is the same only within groups defined by the observed data, then the data are missing at random (MAR)." Since baseline SBP can define observed groups (e.g., high vs. low SBP), and the VGSM confirms that "in MAR, the probability of missing data may depend not only on the covariates in the model but also on observed outcomes," this scenario clearly fits the MAR classification.

Scenario (b): Missingness only depends on baseline diabetes (baseline_dm)

Classification: MAR (Missing at Random) **Explanation**: This scenario follows the same logical framework as scenario (a). Baseline diabetes status is a binary variable that is completely observed for all participants at study entry. Using the VGSM's definition, this cannot be MCAR because "data are said to be missing completely at random (MCAR) if P(Ri=1) does not depend on any of the variables" - yet here missingness explicitly depends on diabetes status. Van Buuren clarifies that this represents MAR because "the probability of being missing is the same only within groups defined by the observed data." Diabetes status creates two clearly defined observed groups (diabetic vs. non-diabetic participants). The VGSM supports this classification, noting that MAR represents "a yet more flexible specification" that "handles a variety of more plausible scenarios" compared to the restrictive MCAR assumption.

Scenario (c): Missingness only depends on SBP at year 1

Classification: NMAR (Not Missing at Random) **Explanation**: This scenario represents a fundamentally different situation from the previous two. Here, missingness depends on the year 1 SBP value itself - the very variable that would be missing. This creates a logical paradox: we cannot observe the year 1 SBP values that determine their own missingness pattern. The VGSM's MCAR definition clearly doesn't apply since missingness depends on a variable. Similarly, van Buuren's MAR criterion - "the probability of being missing is the same only within groups defined by the observed data" - cannot be satisfied because year 1 SBP is unobserved data, not observed data that can define groups. Van Buuren gives the definitive classification: "If neither MCAR nor MAR holds, then we speak of missing not at random (MNAR)" and crucially explains that "MNAR means that the probability of being missing varies for reasons that are unknown to us." Since the year 1 SBP values that drive the missingness mechanism are themselves unobserved, this scenario exemplifies NMAR.

Mean Comparisons

Question: How does mean SBP compare between the complete data and the non-missing subset?

Measurement	Mean SBP
Baseline	134.9 mmHg
Year One (Complete Data)	135.2 mmHg
Year One (Non-Missing Data)	127.5 mmHg

Analysis: - Difference in mean SBP (baseline to year 1, complete data): +0.32 mmHg - Difference using only non-missing values at year 1: -7.34 mmHg - Explanation of the difference: The complete data shows no change in mean SBP, while non-missing data shows a decrease. This occurs because participants with missing data have higher SBP values. When these high values are excluded, the remaining participants show lower mean SBP at year 1.

GEE Analysis

Question: How do GEE estimates compare between the complete data and the non-missing subset?

Complete data GEE results: - Coefficient for factor(visit)1 = 0.32 mmHg - Robust SE = 0.34, z = 0.93, p > 0.05 - 95% CI: (-0.35, 0.98) mmHg - **Note**: Naïve SE (0.53) set aside: naïve standard errors assume independence and underestimate uncertainty in correlated data

Non-missing data GEE results: - Coefficient for factor(visit)1 = -7.34 mmHg
- Robust SE = 0.52, z = -14.23, p < 0.001 - 95% CI: (-8.35, -6.33) mmHg - **Note**: Ignoring naïve SE (0.60) - using robust SE accounts for within-subject correlation

Why robust standard errors: Only robust standard errors are interpreted here. Naïve SEs assume observations are independent, which violates the repeated measures structure. Robust (sandwich) standard errors properly account for within-subject correlation and provide valid inference.

Analysis: - Average change in SBP (complete data): +0.32 mmHg, 95% CI (-0.35, 0.98), not statistically significant - Average change in SBP (non-missing data): -7.34 mmHg, 95% CI (-8.35, -6.33), highly significant ($p < 0.001$) - **Why different**: Complete data includes all participants (even those with missing year 1 data), while non-missing analysis uses only complete cases. The participants with missing data have systematically higher SBP values, creating selection bias when excluded.

Linear Mixed Models

Question: How do mixed model results compare, and how do they contrast with GEE?

Complete data mixed model: - Coefficient for factor(visit)1 = 0.32 mmHg (SE = 0.34, t = 0.93) - 95% CI: (-0.35, 0.98) mmHg

Non-missing data mixed model: - Coefficient for factor(visit)1 = 0.62 mmHg (SE = 0.42, t = 1.47) - 95% CI: (-0.35, 1.54) mmHg

CRITICAL FINDING - Massive Discrepancy Explained: The most important finding is the dramatic difference between GEE and mixed models for non-missing data: - **GEE non-missing**: -7.34 mmHg decrease
- **Mixed model non-missing**: +0.62 mmHg increase

Root Cause Analysis: This discrepancy occurs due to fundamental differences in how the methods handle data:

1. **GEE Approach**: Uses complete cases only. When we specify `nonmiss = alldata[alldata$missing==0,]`, GEE only analyzes participants who have BOTH baseline AND year 1 measurements (n=1,384 participants with 2 observations each).

2. **Mixed Model Approach**: Uses maximum likelihood with all available data. Even with the "non-missing" subset, the mixed model includes ALL participants who have baseline data, regardless of whether they have year 1 data. This includes participants flagged as missing who contribute baseline information.

Analysis: - **Mixed model results comparison**: Complete (+0.32) vs non-missing (+0.62) show similar small positive changes, both non-significant. - **GEE vs Mixed models with missing data**: The -7.34 mmHg vs +0.62 mmHg difference reflects different analytical populations and missing data handling approaches. - **Why they differ**: GEE uses listwise deletion (complete cases only), while mixed models

use all available information under MAR assumptions. Mixed models retain baseline information from all 2,607 participants, while GEE restricts to the 1,384 with complete data, creating substantial selection bias.

Summary and Key Findings

Missing Data Patterns:

46.9% of participants have missing year 1 SBP data. Missing data is strongly associated with baseline characteristics: higher SBP (Cohen's d = 2.01, large effect), higher BMI, and increased diabetes prevalence (OR = 1.55). The large effect size for baseline SBP suggests a systematic, non-random missing data mechanism that is MAR conditional on observed baseline characteristics.

GEE vs Mixed Models - Critical Methodological Differences:

The analysis reveals fundamental differences in missing data handling: - **GEE**: Listwise deletion approach, analyzing only complete cases (n=1,384), showing -7.34 mmHg change - **Mixed Models**: Maximum likelihood under MAR assumptions, utilizing all available data (n=2,607), showing +0.62 mmHg change - **Clinical Significance**: The 7.96 mmHg difference between methods (|-7.34 - 0.62|) exceeds clinically meaningful thresholds for blood pressure changes

Impact of Missing Data on Inference:

Missing data handling dramatically affects conclusions: 1. **Complete data**: No significant change (+0.32 mmHg, $p > 0.05$) 2. **GEE complete cases**: Significant decrease (-7.34 mmHg, $p < 0.001$)
3. **Mixed model MAR**: No significant change (+0.62 mmHg, $p > 0.05$)

The choice between complete case analysis (GEE) versus MAR-based methods (mixed models) leads to opposite conclusions about treatment efficacy, demonstrating the critical importance of appropriate missing data methodology in longitudinal studies.

Methodological Recommendations:

Based on the strong association between missingness and baseline SBP (Cohen's d = 2.01), the MAR assumption appears reasonable. Mixed models provide more valid inference by utilizing all available information and avoiding selection bias inherent in complete case analysis.

Sensitivity Analysis for Missing Data Assumptions:

The large effect size (Cohen's d = 2.01) for baseline SBP association with missingness provides strong evidence supporting the MAR assumption. However, sensitivity analyses should be considered: -

MCAR sensitivity: Complete case analysis (GEE) shows -7.34 mmHg change, suggesting MCAR is violated - **MAR robustness**: Mixed models under MAR show +0.62 mmHg, dramatically different from complete case analysis - **NMAR exploration**: If missingness depended on unobserved year 1 SBP values, pattern mixture models or selection models would be needed - **Recommendation**: The 7.96 mmHg difference between methods suggests MAR assumption is more plausible than MCAR, supporting mixed model approach

References and Theoretical Framework

Primary References:

Vittinghoff, E., Glidden, D.V., Shiboski, S.C., and McCulloch, C.E. (VGSM) Regression Methods in Biostatistics, Chapter 11: "Missing Data" (pages 439-473). This chapter provides the theoretical framework for missing data classifications: - **MCAR (Section 11.2.1.1)**: "Data are said to be missing completely at random (MCAR) if P(Ri=1) does not depend on any of the variables" - **MAR (Section 11.2.1.3)**: "In MAR, the probability of missing data may depend not only on the covariates in the model but also on observed outcomes" - **NMAR**: When missingness depends on unobserved values, creating non-ignorable missing data mechanisms

van Buuren, S. (2018). *Flexible Imputation of Missing Data*, 2nd Edition, Section 1.2: Concepts of MCAR, MAR and MNAR. It provides practical definitions with examples: - **MCAR**: "If the probability of being missing is the same for all cases, then the data are said to be missing completely at random (MCAR)" - **MAR**: "If the probability of being missing is the same only within groups defined by the observed data, then the data are missing at random (MAR)" - **MNAR**: "If neither MCAR nor MAR holds, then we speak of missing not at random (MNAR)...MNAR means that the probability of being missing varies for reasons that are unknown to us"

Missing Data Theory: Little, R.J.A. and Rubin, D.B. (2019). Statistical Analysis with Missing Data, 3rd Edition. Provides comprehensive theoretical foundation for missing data mechanisms and inference methods.

GEE Methodology: Liang, K.Y. and Zeger, S.L. (1986). Longitudinal data analysis using generalized linear models. Biometrika, 73(1), 13-22. Naïve standard errors assume independence, which are inappropriate for correlated data. Robust (sandwich) standard errors provide valid inference under mild regularity conditions.

Mixed Models Theory: Laird, N.M. and Ware, J.H. (1982). Random-effects models for longitudinal data. Biometrics, 38(4), 963-974. Mixed models handle missing data through maximum likelihood estimation under MAR assumptions, providing more efficient estimates than complete case analysis when missingness is related to observed covariates.

Effect Size Interpretation: Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences, 2nd Edition. Cohen's d values: 0.2 (small), 0.5 (medium), 0.8 (large effect sizes).